

$$(ABC) + (\bar{A}BC)$$

1 1 1

$$ABC = m_7$$

0 1 1

$$\bar{A}BC = m_3$$

1 0 1

$$A\bar{B}C = m_5$$

		BC			
		00	01	11	10
A	0	0	1	3	2
	1	4	5	7	6

$$= m_7 + m_3 + m_5$$

$$= \sum m(3, 5, 7)$$

Minterms and Maxterms

Variables			Minterms
A	B	C	
0	0	0	$\bar{A}\bar{B}\bar{C} = m_0$
0	0	1	$\bar{A}\bar{B}C = m_1$
0	1	0	$\bar{A}B\bar{C} = m_2$
0	1	1	$\bar{A}BC = m_3$
1	0	0	$AB\bar{C} = m_4$
1	0	1	$AB\bar{C} = m_5$
1	1	0	$ABC = m_6$
1	1	1	$ABC = m_7$

Example

Design a logic circuit that produces 1 when the input BCD digit is an odd number.

A B C D

0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

0
1
0
1
0
1
0
1
0
1

		CD			
		00	01	11	10
AB	00	0	1	1	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

= D

		CD			
		00	01	11	10
AB	00	1			
	01			1	1
	11	1		1	
	10	1	1		

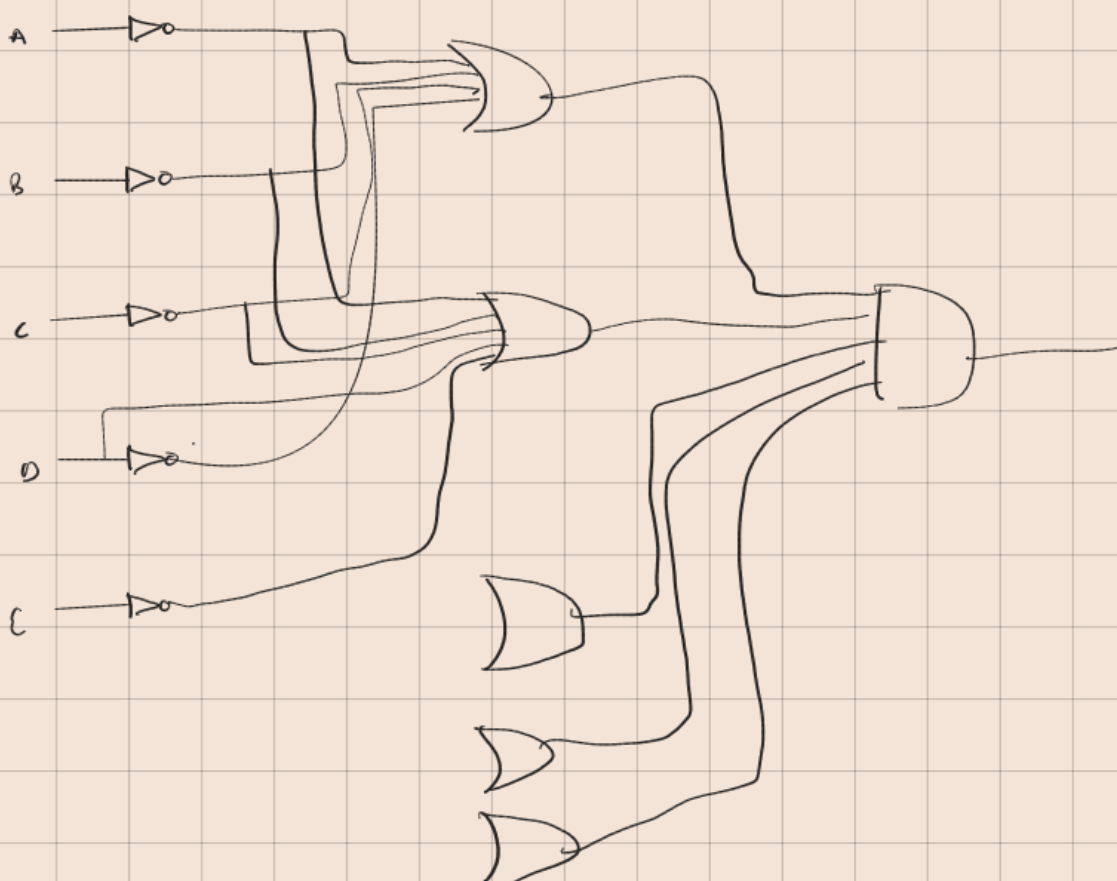
Five people judge a certain competition. the vote of each is indicated by '1' (pass) or '0' (fail) on a signal line. The five signal lines form an input to the logic circuit. The rule of competition allow only one dissenting vote. The logic circuit is to have 2 outputs, x and y. If the vote is 4-1 or 5-0 to pass, $xy = 11$. If the vote is 4-1 or 5-0 to fail, $xy = 00$. If the vote 3-2 or 2-3, $xy=10$. Design the logic circuit for this problem?

$$\begin{array}{r} \underline{11110} + \underline{11111} = \underline{11} \\ \underline{00001} + \underline{00000} = \underline{00} \\ \underline{11100} + \underline{11000} = \underline{10} \end{array}$$

CE	000	001	011	010	110	111	101	100
AB	00	01	11	10	00	01	11	10
00	0	1	1	2	2	7	5	4
01	9	9	11	10	14	15	13	12
11	24	25	27	28	30	31	29	28
10	16	17	19	18	22	23	21	20

CE	000	001	011	010	110	111	101	100
AB	00	01	11	10	00	01	11	10
00	0	1	1	2	2	7	5	4
01	9	9	11	10	14	15	13	12
11	24	25	27	28	30	31	29	28
10	16	17	19	18	22	23	21	20

$$BC' + BCE + A'B'O'E' + A'B'D'E' + AB'CO'E + AB'CO'E$$



Example 2

$$F(A, B, C, D, E) = \Sigma m(4, 5, 6, 7, 20, 21, 22, 23) + d(0, 1, 2, 3)$$

CE	000	001	011	010	110	111	101	100
AB								
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

$$X + \overline{X} \cdot Y = X + Y$$

$$\overline{A} + A \cdot C =$$

$$X = \overline{A} \quad \overline{X} = A =$$

$$Y = C$$

$$X + \overline{X} \cdot Y$$

$$(\overline{A}) + (\overline{A}) \cdot C$$

$$(ab) + (\bar{a}cd) + (a\bar{b}\bar{d})$$

$$(ab) + (\bar{a}cd) + (db\bar{d})$$

$$(\bar{a}b) + (\bar{a}cd) + 0$$

$$\bar{a}(b+cd)$$

$$\bar{a}(b+cd) \quad \therefore \overline{xy} = \bar{x} + \bar{y}$$

$$\bar{a} + \overline{(b+cd)}$$

$$a + \overline{(b+cd)} \quad \therefore \overline{x+y} = \bar{x} \cdot \bar{y}$$

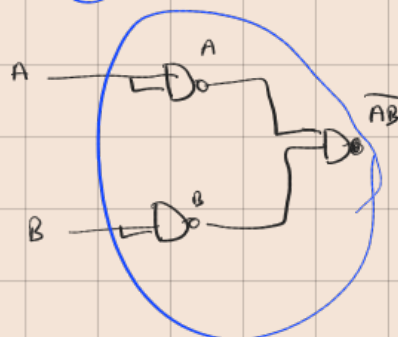
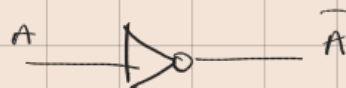
$$a + \bar{b} \cdot \bar{c}\bar{d}$$

$$a + \bar{b} + (\bar{c} + \bar{d})$$

$$a + \bar{b}\bar{c} + \bar{b}\bar{d}$$

A	B	C	D	F	
0	0	0	0	0	0
0	0	0	1	0	
0	0	1	0	1	1
0	0	1	1	1	
0	1	0	0	0	D
0	1	0	1	1	
0	1	1	0	0	0
0	1	1	1	1	
1	0	0	0	0	0
1	0	0	1	0	
1	0	1	0	1	1
1	0	1	1	1	
1	1	0	0	1	D'
1	1	0	1	0	
1	1	1	0	1	D'
1	1	1	1	0	0

A	B	C	F
0	0	0	
0	0	1	
0	1	0	D
0	1	1	
1	0	0	
1	0	1	
1	1	0	D'
1	1	1	



$$F(A,B,C,D,E) = \sum m(4, 5, 6, 7, 20, 21, 22, 23) + \underbrace{d(0, 1, 2, 3)}$$

	a	b	c	d	e	Σ
0	0	0	0	0	0	X
1	0	0	0	0	1	X
2	0	0	0	1	0	X
3	0	0	0	1	1	X
4	0	0	1	0	0	1
5	0	0	1	0	1	1
6	0	0	1	1	0	1
7	0	0	1	1	1	1
8	0	1	0	0	0	0
9	0	1	0	0	1	0
10	0	1	0	1	0	0
11	0	1	0	1	1	0
12	0	1	1	0	0	0
13	0	1	1	0	1	0
14	0	1	1	1	0	0
15	0	1	1	1	1	0
16	1	0	0	0	0	0
17	1	0	0	0	1	0
18	1	0	0	1	0	0
19	1	0	0	1	1	0
20	1	0	1	0	0	1
21	1	0	1	0	1	1
22	1	0	1	1	0	1
23	1	0	1	1	1	1
24	1	1	0	0	0	0
25	1	1	0	0	1	0
26	1	1	0	1	0	0
27	1	1	0	1	1	0
28	1	1	1	0	0	0
29	1	1	1	0	1	0
30	1	1	1	1	0	0
31	1	1	1	1	1	0

a	b	c	d	Σ
0	0	0	0	X
1	0	0	0	X
2	0	0	1	X
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

a	b	c	Σ
0	0	0	X
1	0	0	1
2	0	1	0
3	0	1	0
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	0

$$A + (A \cdot B) = A$$

$$ABC\overline{D} + D = D + ABC$$

$$\begin{aligned} \boxed{X + \overline{X}Y} &= X + Y \\ \boxed{D + \overline{D}ABC} &= D + ABC \quad (\text{proved}) \end{aligned}$$

$$AC\overline{B} + CAD\overline{B} = A\overline{B}C$$

$$\underline{A\overline{B}C} + \underline{A\overline{B}C\overline{D}} = A\overline{B}C$$

$$A + \overline{T} + \overline{W} + \overline{A} + X = 1$$

$$A + \overline{A} + \overline{T} + \overline{W} + X = 1$$

$$(1 + \overline{T}) + \overline{W} + X = 1$$

$$(1 + \overline{W}) + X = 1$$

$$(1 + X) = 1$$

$$1 = 1 \quad \text{proved}$$

$$X\overline{Y}\overline{Z} + \overline{X} = \overline{X} + \overline{Y}Z$$

$$\begin{aligned} \boxed{X} + \boxed{\overline{X}Y} &= X + Y \\ \boxed{\overline{X}} + \boxed{X\overline{Y}Z} &= \overline{X} + \overline{Y}Z \end{aligned}$$

$$\overline{G}\overline{F}\overline{H} \quad \overline{H}\overline{G}\overline{F} = \overline{F}\overline{H}\overline{G}$$

$$\overline{C}\overline{A}\overline{B} + AB = AB + C$$

$$X + \overline{X}Y = X + Y$$

$$AB + \overline{A}\overline{B}C = AB + C$$

